

SANDESH SIGDEL

Full-Stack Developer | MERN | Pokhara, Nepal
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SUMMARY

Software Engineering graduate and self-taught full-stack developer who ships production systems end to end. Built and deployed a real-time blood-donation platform, covering JWT authentication, a role-based authorization model, WebSocket notifications, a weighted donor-matching algorithm, and a hardened API. Currently deepening cloud knowledge (AWS Certified Cloud Practitioner in progress).

TECHNICAL SKILLS

Languages: JavaScript (ES6+), HTML5, CSS3, SQL (basic)

Frontend: React 18, React Router, Vite, Context API, Responsive Design, Axios

Backend: Node.js, Express.js, RESTful API Design, Socket.IO / WebSockets, JWT, bcrypt

Databases: MongoDB, Mongoose, Schema Design, Indexing, Aggregation Pipelines

Tools & Cloud: Git, GitHub, Postman, VS Code, Vercel, MongoDB Atlas, Cloudinary, AWS (learning)

Practices: Authentication & Authorization, API Security (Helmet, rate limiting, input sanitization), Query Optimization, Agile Collaboration

PROJECTS

RaktaSetu: Real-Time Blood Donation Platform

React, Node.js, Express, MongoDB, Socket.IO

Full-stack MERN platform connecting blood donors with recipients during medical emergencies.

- Architected a full-stack MERN application with a React SPA on Vercel and an Express API serving both REST and WebSocket traffic from a single Node process.
- Built a real-time notification system using Socket.IO rooms that pushes matched blood requests directly to each donor's private channel, eliminating client polling and reducing notification latency to near-instant.
- Designed a weighted donor-ranking algorithm scoring proximity (Haversine distance), availability, donation eligibility, and verification status, replacing a naive radius-expansion search that required multiple sequential queries.
- Implemented three-tier authorization: JWT authentication, role-based access control, and an admin verification gate for hospital accounts, ensuring only vetted institutions can publish inventory or requests.
- Optimized MongoDB performance with compound indexes matching query filter order and a single aggregation pipeline (\$unwind/\$match/\$group) that eliminated an N+1 query pattern across the donor candidate pool.
- Hardened the API against common attacks: bcrypt password hashing, NoSQL injection sanitization, tiered rate limiting, IDOR ownership checks, and CSPRNG-generated hashed OTPs with TTL and attempt limits for password reset.

Imposter: Hidden-Role Multiplayer Web Game

HTML, CSS, JavaScript

- Built and deployed a browser-based social deduction game with real-time role assignment and multi-player game state handled entirely client-side in vanilla JavaScript.
- Implemented full game logic from scratch, including role distribution, round management, and voting, without frameworks or external libraries, then handled deployment, hosting, and on-page SEO optimization end to end.

Happy Paws: Pet Food E-Commerce Storefront

React, JavaScript, CSS

- Built and deployed a live, fully responsive React storefront for pet food products, structured around reusable components and client-side routing for fast navigation between catalog and product pages.
- Managed product catalog and user interaction state with React hooks, and handled build, hosting, and deployment independently.

EDUCATION

Bachelor of Software Engineering

2021 to 2025

School of Engineering, Pokhara University

CERTIFICATIONS & LANGUAGES

Certifications: Google Digital Marketing

AWS Certified Cloud Practitioner (In Progress, Expected Aug 2026)

Languages: English (Fluent) | Nepali (Native) | Hindi (Conversational)